CLINICAL · EAR

Ear signs and eponyms at a glance

Otology is full of names. This page gathers the signs, syndromes and anatomical landmarks that examiners love, each explained in a line or two and pointing to the topic where it belongs. Use it for revision the night before a viva.

AT A GLANCE

- A one-stop revision sheet of signs, syndromes and landmarks.
- Each links to the topics that explain it.

1 CHAPTER ONE Signs

What you see or elicit.

1.1 Abscesses around the mastoid

1.1.1 Bezold abscess

Pus breaking through the medial side of a pneumatised mastoid tip and collecting under the sternocleidomastoid. See Mastoiditis.

1.1.2 Citelli abscess

Pus tracking from the tip along the posterior belly of digastric, behind the mastoid or in the digastric triangle.

1.1.3 Luc abscess

A meatal abscess: pus from the antrum tracking under the periosteum of the bony canal.

1.2 Signs of intracranial spread

1.2.1 Griesinger's sign

Oedema over the mastoid emissary vein in lateral sinus thrombosis.

1.2.2 Tobey-Ayer test

Compressing the jugular vein on the healthy side raises CSF pressure, on the thrombosed side does not; historical.

1.2.3 Crowe-Beck test

Compressing the healthy jugular vein engorges the retinal veins in lateral sinus thrombosis; historical.

1.2.4 Gradenigo's triad

Ear discharge, deep pain behind the eye and a sixth nerve palsy in petrous apicitis.

1.3 Labyrinth

1.3.1 Hennebert's sign

A positive fistula test with an intact drum: congenital syphilis, Ménière's disease, superior canal dehiscence.

1.3.2 Tullio phenomenon

Vertigo triggered by loud sound: superior canal dehiscence, perilymph fistula, hydrops.

1.4 Otosclerosis

1.4.1 Schwartze sign

A red blush on the promontory through the drum, from an active, vascular focus.

1.4.2 Carhart's notch

A dip of bone conduction maximal at 2000 Hz, mechanical, disappearing after stapes surgery.

1.4.3 Paracusis Willisii

Hearing better in noisy surroundings.

1.5 Tumours

1.5.1 Brown's sign

A glomus tumour pulsates, then blanches, as pressure in the canal is raised.

1.5.2 Rising sun

A red glomus tumour seen rising from the floor of the middle ear behind the drum.

1.5.3 Hitselberger's sign

Reduced sensation of the posterior canal wall, from pressure on the sensory fibres of the facial nerve by a vestibular schwannoma.

1.6 Facial nerve

1.6.1 Bell's phenomenon

The eye rolls up and out when the patient tries to close it.

1.7 Mastoiditis

1.7.1 Reservoir sign

The canal refills with pus straight after it is mopped.

1.7.2 Light-house sign

Pulsatile discharge through a pars tensa perforation.

2 CHAPTER TWO Syndromes

Named collections of findings.

2.1 Facial nerve

- **Ramsay Hunt:** herpes zoster oticus with facial palsy, often with hearing loss and vertigo.
- **Melkersson-Rosenthal:** recurrent facial palsy, swelling of the lips and a fissured tongue.
- **Heerfordt:** sarcoidosis with uveitis, parotid swelling and facial palsy.
- **Möbius:** congenital bilateral facial palsy with sixth nerve palsy.
- **Bogorad (crocodile tears):** tearing while eating after facial nerve regeneration.
- **Frey:** gustatory sweating after parotid surgery.

- **Bannwarth:** facial palsy in Lyme meningoradiculitis.

2.2 Inner ear

- **Cogan:** interstitial keratitis with audiovestibular loss.
- **Vogt-Koyanagi-Harada:** uveitis, alopecia, white hair and deafness.
- **Lermoyez:** hearing improves with the vertigo attack.
- **Tumarkin's otolithic crisis:** sudden drop attacks in Ménière's disease.
- **Mal de débarquement:** a rocking sensation persisting after travel by sea or air.
- **van der Hoeve:** osteogenesis imperfecta, blue sclerae and deafness.

2.3 Pain

- **Costen:** a faulty bite with earache, fullness, tinnitus and dizziness.
- **Eagle:** an elongated styloid process with throat and ear pain.

2.4 Jugular foramen

- **Vernet:** palsy of the ninth, tenth and eleventh nerves.
- **Collet-Sicard:** ninth to twelfth nerves.
- **Villaret:** ninth to twelfth nerves with Horner syndrome.

3 CHAPTER THREE Spaces and landmarks

Named anatomy.

3.1 Middle ear and mastoid

- **Prussak's space:** between the pars flaccida and the neck of the malleus; where attic cholesteatoma begins.
- **Trautmann's triangle:** bounded by the bony labyrinth, the sigmoid sinus and the superior petrosal sinus (dura); a route to the posterior fossa.

- **Citelli's angle:** the sinodural angle, between the sigmoid sinus and the middle fossa dura.
- **MacEwen's triangle:** the suprameatal triangle, marking the antrum.
- **Körner's septum:** a persistent petrosquamous plate in the mastoid, which can be mistaken for the floor of the antrum.

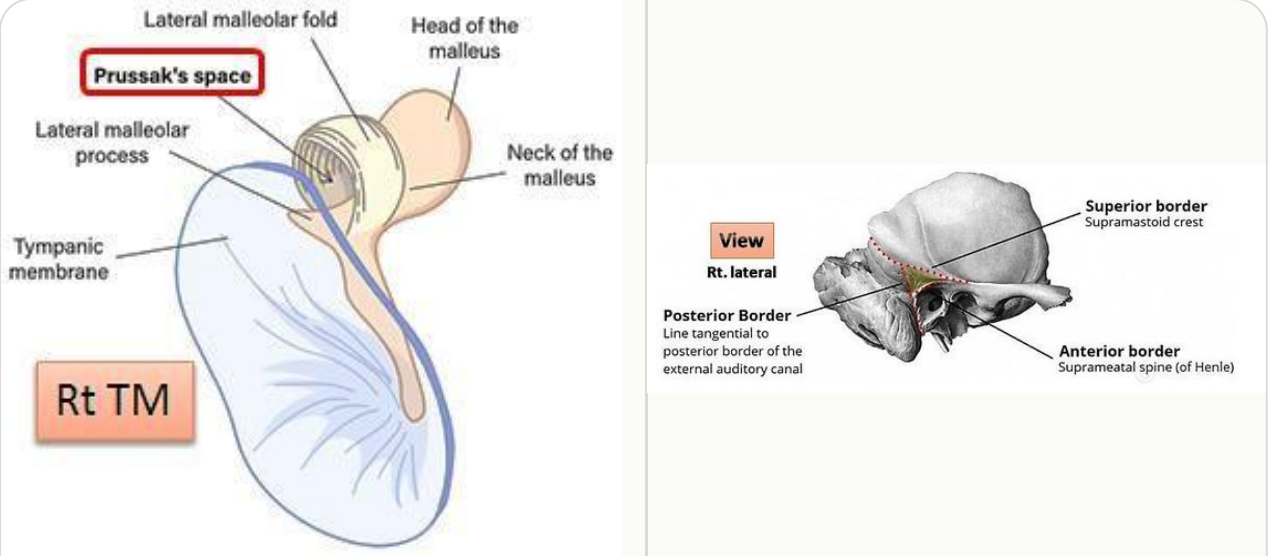


FIGURE 1 Prussak's space, and MacEwen's triangle.
Anatomy atlas · ENT Steps p. 54, ENT Steps p. 56

3.2 Canal

- **Foramen of Huschke:** a defect in the anteroinferior bony canal in children, a route between the canal and the parotid.
- **Fissures of Santorini:** clefts in the cartilaginous canal through which infection passes to the parotid.

3.3 Skull base

- **Dorello's canal:** where the sixth nerve passes under the petroclinoid ligament at the petrous tip.
- **Meckel's cave:** the trigeminal ganglion on the petrous apex.
- **Hyrtl's fissure:** an infant fissure from the hypotympanum to the subarachnoid space.

3.4 Nerves

- **Arnold's nerve:** the auricular branch of the vagus; explains the ear cough reflex.
- **Jacobson's nerve:** the tympanic branch of the glossopharyngeal; origin of glomus tympanicum and route for referred pain.

---

4 CHAPTER FOUR For the examination

Using this page.

HOW TO USE THE NAMES

- Say what the sign is, then what it means, then which disease it points to.
- Know which tests are historical (Tobey-Ayer, Crowe-Beck) and what has replaced them (imaging).
- Link each landmark to the operation where it matters.

Written for this app from Dhingra ch. 1, 12, 14, 17 and K. J. Lee ch. 1, 13, 16. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.